

	[8Hrs]
Unit II	AI Learning Models: Concept of learning, learning automation, Knowledge-Based Classification Inductive Learning, Deductive Learning: Feedback Based Classification Unsupervised Learning, Supervised Learning, Semi-supervised learning and Reinforcement Learning., Deep Learning ANN and CNN [7Hrs]
Unit III	Cybernetic View of Robot Cognition and Perception: Introduction to the Model of Cognition, Visual Perception, Visual Recognition, Machine Learning, Soft Computing Tools and Robot Cognition. [7 Hrs]
Unit IV	Heuristic Search: Heuristic Functions, Hill Climbing, Variable Neighborhood Descent, Beam Search, Tabu Search, Optimal Search: A * algorithm, Iterative Deepening A*, Recursive Best First Search, Pruning the CLOSED and OPEN Lists. Problem Decomposition: Goal Trees, Rule Based Systems, Rule Based Expert Systems. Planning: STRIPS, Forward and Backward State Space Planning, Goal Stack Planning, Plan Space Planning, A Unified Framework For Planning. Constraint Satisfaction: N-Queens, Constraint Propagation, Backtracking and Look ahead Strategies [8 Hrs]
Unit V	Robotics: Fundamentals, path Planning for Point Robot, Sensing and mapping for Point Robot, Mobile Robot Hardware, Non Visual Sensors like: Contact Sensors, Inertial Sensors, Infrared Sensors, Sonar, Radar, laser Rangefinders, Biological Sensing. Robot System Control: Horizontal and Vertical Decomposition, Hybrid Control Architectures, Middleware, High-Level Control, Human-Robot Interface. [8 Hrs]